

Business Problem Statement

A top retail firm wants to understand the shopping habits of its customers to lead the sales growth, improve customer satisfaction and foster long-time customer loyalty. The management is interested in what they need to know about the key factors that affect purchasing, including discounts, customer reviews, seasonality and preferred payment methods. They also wish to uncover trends that can be used to understand customer retention and repeat purchases.

The aim of this analysis is to derive meaningful trends and consumer behaviour patterns from the consumer's shopping behaviour, which can help in taking data-driven business decisions.

“How can the company leverage consumer shopping data to identify trends, improve customer engagement, and optimize marketing and product strategies?”

Customer Shopping Behavior Analysis - Report

1. Project Overview

This project analyzes customer trends and patterns using data from 3,900 purchases for various categories of products of an organization. The aim is to reveal insights into spending patterns, customer segments, products references, and customer subscription status to guide strategic business decisions.

2. Dataset Structure

- **Rows:** 3,900
- **Columns:** 18
- **Customer Demographics** (age, gender, location, subscription status)
- **Purchase details** (item purchased, category, purchase amount, season, size, color)
- **Shopping behavior** (discount applied, promo code used, previous purchases, frequency of purchases, review rating, shipping type)

- **Missing Data:** We found 37 missing values in the Review Rating column

3. Exploratory Data Analysis using Python

We began with data preparation and cleaning:

- **Data Loading:** Imported the dataset using pandas into **VSCode**
- **Initial Exploration:** Checked the basic structure and overview of the dataset using **df.info()** and **df.describe()**

	customer_id	age	purchase_amount	review_rating	previous_purchases
count	3900.000000	3900.000000	3900.000000	3900.000000	3900.000000
mean	1950.500000	44.068462	59.764359	3.750051	7.121026
std	1125.977353	15.207589	23.685392	0.713590	5.598860
min	1.000000	18.000000	20.000000	2.500000	1.000000
25%	975.750000	31.000000	39.000000	3.100000	4.000000
50%	1950.500000	44.000000	60.000000	3.800000	6.000000
75%	2925.250000	57.000000	81.000000	4.400000	8.000000
max	3900.000000	70.000000	100.000000	5.000000	30.000000

- **Missing Data Handling:** Checked the missing values in Review Rating column and replaced them with median rating of each product category.
- **Column Standardization:** Renamed all columns to snake case so as to increase readability and ease of usage.
- **Column Addition:**
 - Created **age_group** column by grouping the customer ages into four segments.
 - Created **purchase_frequency_days** column from purchase data.
- Compared **discount_applied** and **promo_code_used** columns to see if they were redundant, and then dropped out **promo_code_used** column
- **Database Integration:** Connected Python script to **PostgreSQL**, and loaded the cleaned DataFrame into the database for SQL analysis.

4. Data Analysis using SQL

We performed structured analysis in **PostgreSQL** to answer key business questions:

1. **Revenue by gender** – Compared total revenue generated by male vs. female customers.

	gender text	revenue numeric	count bigint
1	Female	75191	1248
2	Male	157890	2652

- High-spending discount users** – Identified customers who used discounts but still spent above the average purchase amount.

	customer_id bigint	purchase_amount bigint
1	2	64
2	3	73
3	4	90
4	7	85
5	9	97
6	12	68
7	13	72
8	16	81
9	20	90
10	22	62

- Top 5 products by Rating** – Found products with the highest average review ratings.

	item_purchased text	avg_product_rating numeric
1	Gloves	3.86
2	Sandals	3.84
3	Boots	3.82
4	Hat	3.80
5	Skirt	3.78

- Shipping Type Comparison** – Compared average purchase amounts between Standard and Express shipping.

	shipping_type text	round numeric
1	Standard	58.46
2	Express	60.48

- Subscribers vs. Non-Subscribers** – Compared average spend and total revenue across subscription status.

	subscription_status text	number_of_customers bigint	average_spend numeric	revenue numeric
1	Yes	1053	59.492	62645.000
2	No	2847	59.865	170436.000

6. **Discount-Dependent Products** – Identified 5 products with the highest percentage of discounted purchases.

	item_purchased text	percentage numeric
1	Hat	50.00
2	Sneakers	49.66
3	Coat	49.07
4	Sweater	48.17
5	Pants	47.37

7. **Customer Segmentation** – Classified customers into New, Returning, and Loyal segments based on purchase history.

	segment text	Number of Customers bigint
1	Loyal	886
2	New	225
3	Returning	2789

8. **Top 3 Products per Category** – Listed the most purchased products within each category.

	item_rank bigint	category text	item_purchased text	total_orders bigint
1	1	Accessori...	Jewelry	171
2	2	Accessori...	Sunglasses	161
3	3	Accessori...	Belt	161
4	1	Clothing	Blouse	171
5	2	Clothing	Pants	171
6	3	Clothing	Shirt	169
7	1	Footwear	Sandals	160
8	2	Footwear	Shoes	150
9	3	Footwear	Sneakers	145
10	1	Outerwear	Jacket	163
11	2	Outerwear	Coat	161

- 9. Repeat Buyers & Subscriptions** – Checked whether customers with >5 purchases are more likely to subscribe.

	subscription_status text	repeat_buyers bigint
1	No	1567
2	Yes	584

- 10. Revenue by Age Group** – Calculated total revenue contribution of each age group.

	age_group text	total_revenue numeric
1	Young Adult	62143
2	Middle-aged	59197
3	Adult	55978
4	Senior	55763

- 11. Product demands according to season** - Analysed revenue and order volume across product category and season.

	category text	season text	revenue numeric	orders bigint
1	Accessori...	Fall	19874	324
2	Accessori...	Summer	19028	312
3	Accessori...	Winter	18291	303
4	Accessori...	Spring	17007	301
5	Clothing	Spring	27692	454
6	Clothing	Winter	27274	448
7	Clothing	Fall	26220	427
8	Clothing	Summer	23078	408
9	Footwear	Spring	9555	163
10	Footwear	Summer	9393	160
11	Footwear	Fall	8665	136
12	Footwear	Winter	8480	140
13	Outerwear	Fall	5259	88
14	Outerwear	Winter	4562	80
15	Outerwear	Spring	4425	81
16	Outerwear	Summer	4278	75

5. Dashboard in Power BI

Finally, we built an interactive dashboard in **Power BI** to present the insights visually.



6. Business Recommendations

- Promote benefits for subscribers so as to attract more subscriptions.
- Reward repeat buyers with coupons, discounts, scratch cards so that they move to the 'Loyal' segment.
- Best-selling products need to be positioned and highlighted in campaigns
- Target high-revenue age groups and express-shipping users more
- Discount policies should be managed to boost sales while maintaining margins.
- Prioritize Outerwear inventory and marketing spend ahead of Fall, while maintaining steady-state stocking for Accessories, Clothing, and Footwear year-round.